Criterion Conflict in Segmentation Reliability Evaluation: Empirical Bounds and Endpoint-Specific Scoring

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Code and derived results

Abstract

Segmentation reliability is evaluated for several distinct purposes, including failure detection and model selection. We examine whether a single score supports these purposes across five six-model panels and 59,880 model–image records from COCO, PASCAL VOC, ADE20K, and Cityscapes. For seven declared criteria, an empirical pairwise-conflict bound ranges from 10.6% to 18.6%: every complete ranking of the evaluated models must disagree with at least this fraction of the non-tied criterion preferences. A mixed-failure score selected on COCO and VOC improves failure-detection AUROC over the strongest single metric on four panels by 0.026–0.068, but trails severity-weighted arithmetic by 0.124 on Cityscapes. Its advantage also changes with the false-positive threshold. We then evaluate a frozen endpoint-routing protocol, with score formulas selected on COCO and VOC before transfer evaluation. Compared with applying the mixed-failure score to every endpoint, routing improves mean policy-rank agreement by 0.318, operational-rank agreement by 0.138, and severe-miss AUROC by 0.237 in the unweighted mean over panels. The gains are positive on all five panels for these three endpoints, with paired uncertainty estimated by image-cluster bootstrap. The panels are fixed, partially overlapping evaluations, not independent samples of future datasets. These results characterize criterion conflict within the evaluated panels and motivate reporting scores against declared endpoints; they do not establish a universally superior metric.

1. Introduction

Segmentation metrics summarize different properties of a prediction. Mean intersection over union (mIoU) measures region overlap after pixel aggregation; Boundary IoU emphasizes contours; tail statistics emphasize poorly covered components; and false-positive scores penalize unsupported predictions. A score that separates failed from non-failed images need not rank models in the same order as a declared policy or operational utility. Treating these uses as interchangeable can obscure the decision a metric is intended to support.

We study this issue through pairwise preferences over complete six-model image groups. For each image and model pair, seven criteria may agree, tie, or disagree. The minority count among non-tied preferences gives a lower bound on the disagreements incurred by any single ordering for that pair. Aggregating these counts yields an empirical bound for the evaluated criteria and models; it is not an impossibility theorem for other criteria, models, or tasks. Across five panels, the estimated bound is 10.6–18.6%. It remains nonzero in model-subset, image-subsample, and alternative tie-rule analyses.

We next examine transfer of a mixed-failure score selected on COCO and VOC. This score improves failure-detection AUROC on four evaluation panels, but its AUROC is 0.124 below severity-weighted arithmetic on Cityscapes. It also reduces policy and operational rank agreement relative to endpoint-specific scores. The direction of its AUROC advantage changes across false-positive thresholds. These findings motivate evaluation protocols that state the target endpoint rather than treating failure detection as a proxy for every decision.

To test such a protocol, we select five score formulas on COCO and VOC, freeze them, and assign each to a declared endpoint before evaluation on ADE20K and Cityscapes. The protocol performs no learned fusion and does not estimate deployment utility; it compares endpoint-specific scores with reuse of the mixed-failure score. The experiments make three contributions: (1) an empirical, criterion-dependent bound on scalar-ranking disagreement; (2) a five-panel transfer study that documents both gains and negative transfer for the mixed-failure score; and (3) an evaluation of frozen endpoint routing, with uncertainty and sensitivity analyses

across datasets, architectures, model subsets, thresholds, and tie rules.

The study is limited to the declared criteria, panels, and endpoints. Its results support reporting the decision endpoint and its associated score alongside conventional segmentation metrics; they do not establish that one score is optimal for deployment.

2. Related Work

Segmentation metrics and problem-aware validation. Mean intersection over union is widely used for semantic segmentation, but it compresses class- and pixel-level error structure into one aggregate. Boundary IoU is designed to increase sensitivity to contour errors, particularly where standard region overlap is insensitive to object-scale boundary deviations [4]. More broadly, validation metrics can produce misleading comparisons through implementation, aggregation, ranking, or interpretation choices. Metrics Reloaded recommends selecting complementary measures from an explicit description of the target problem, and its companion analysis organizes recurrent metric-related pitfalls [17, 18]. We operationalize one such concern: different correctly computed criteria can induce incompatible model preferences. Our finite-panel bound quantifies that incompatibility for the criteria declared here; it does not identify a universally optimal metric set.

Reliability and uncertainty in segmentation. Semantic-segmentation reliability includes distinct tasks such as robustness, calibration, misclassification detection, and out-of-distribution detection. A broad architecture study found that architectural progress on robustness does not imply uniform improvement in uncertainty-based reliability [8]. ValUES advocates controlled validation across uncertainty applications and emphasizes that score aggregation is itself a consequential design choice [12]. Our scope is narrower: we compare fixed per-image quality scores for model ranking and failure discrimination. We do not estimate pixelwise predictive calibration, detect out-of-distribution inputs, or claim that these tasks are interchangeable.

Safety- and relevance-aware evaluation. Driving-segmentation metrics can account for spatial error structure [6]. Relevance-weighted IoU emphasizes corner-case pixels [2], while Safe mIoU represents class-confusion costs through a hierarchy [20]. These approaches make task priorities explicit. We similarly declare severity weights and operational thresholds, but treat them as experimental assumptions rather than validated harm models. We report policy agreement, operational agreement, severe misses, and false positives separately so that one endpoint cannot stand in for all others.

Tail risk, selection, and endpoint choice. Conditional value at risk (CVaR) summarizes the worst part of a loss distribution [19]; minimum component coverage provides an extreme component-level statistic. Selective-prediction methods instead evaluate risk as a function of retained coverage and can provide guarantees under their assumptions [10]. We evaluate neither abstention nor risk–coverage guarantees. Our question is whether failure-detection discrimination and model-ranking agreement can be treated as the same endpoint. The results show why that implication must be tested rather than assumed: a mixed-failure AUROC gain does not guarantee better policy or operational ranking.

Architectural diversity. The evaluated panels include FCN [16], DeepLabV3 [3], MobileNetV3 [11], SegFormer [22], UPerNet [21], ConvNeXt [15], Swin [14], BEiT [1], and Mask2Former [5]. This permits a limited cross-architecture analysis, but each panel contains only six evaluated checkpoints. It is not a broad sample of the architecture space.

3. Evaluation Framework

3.1. Component scores and endpoint targets

For image x and model m, let eligible target component i have coverage $c_i \in [0,1]$, provenance $p_i \in [0,1]$, and externally assigned severity $s_i > 0$. The candidate score set includes foreground IoU, Boundary IoU, minimum component coverage, severity-weighted arithmetic and harmonic means, CVaR quality, false-positive quality $q_{\mathrm{fp}} = 1 - r_{\mathrm{fp}}$, and the geometric component score

$$q_{\mathrm{comp}} = \exp\left(\sum_i \frac{s_i^\alpha}{\sum_j s_j^\alpha} \log[\epsilon + (1-\epsilon)c_i p_i]\right)(1-r_{\mathrm{fp}})^\gamma. \tag{1}$$

The implementation sets $\alpha = 2$, $\epsilon = 10^{-6}$, and $\gamma = 1$. Scores are oriented so that larger values indicate better quality. This score is one candidate in the comparison, not a universal recommendation.

The declared policy utility is one minus the largest severity-normalized component risk. Operational utility is the minimum of policy utility and q_{fp}. An image is labelled as containing a severe miss if an eligible critical component has coverage below 0.5. A high-false-positive image has $r_{\mathrm{fp}} \geq 0.25$. Their union defines the primary mixed-failure endpoint. These thresholds and severity weights are experimental operationalizations, not causal labels of harm or validated deployment costs.

All targets and score components are computed against reference annotations; these are retrospective, annotation-dependent quality assessments, not annotation-free online reliability estimates. The false-positive endpoint is particularly direct: $h_{\mathrm{fp}} = 1 - r_{\mathrm{fp}}$, so its error score is $1 - h_{\mathrm{fp}} = r_{\mathrm{fp}}$, while

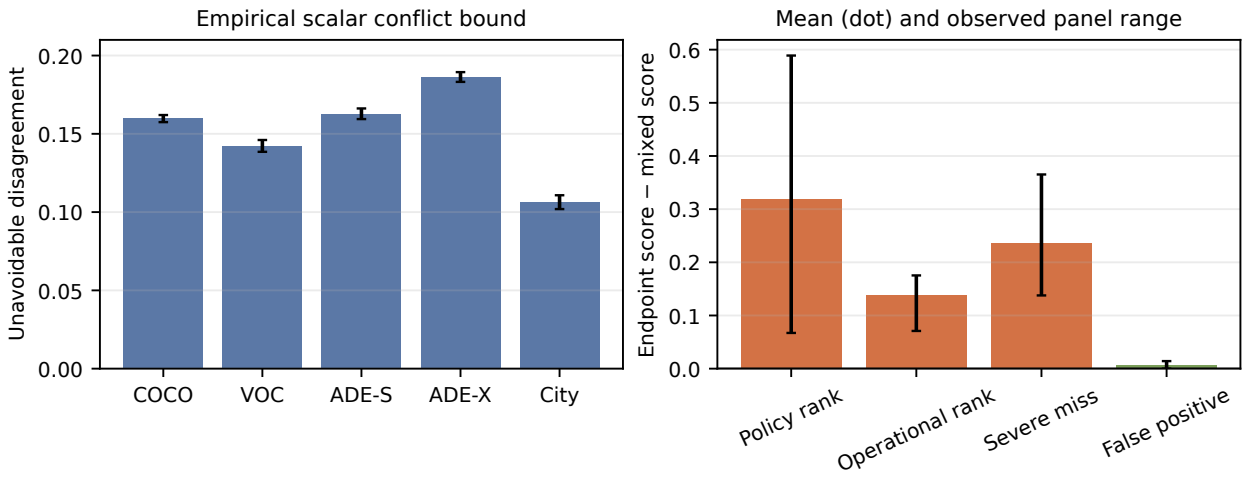


Figure 1. Left: empirical lower bound on pairwise disagreement for any scalar ordering over seven declared criteria (95% image-bootstrap confidence intervals). Right: unweighted mean change from endpoint-specific scoring relative to reuse of the mixed-failure score; whiskers show the observed range across the five panels, not a confidence interval. Panel-specific paired image-bootstrap intervals are shown in Fig. 3. ADE-S is the SegFormer panel; ADE-X is the cross-architecture panel.

its label is $\mathbb{1}[r_{\text{fp}} \geq 0.25]$. Whenever both label classes occur, this construction yields AUROC 1 by definition. We treat that result as a check of alignment with a directly constructed endpoint, not as evidence of general failure-detection ability.

Policy and operational targets are evaluated as rankings of six models within each image: we compute Spearman rank correlation for each image and average across images with non-degenerate ranks. Severe-miss, false-positive, and mixed-failure targets are binary image labels evaluated with AUROC using $1 - h_e$ as the error score. Rank agreement and AUROC therefore estimate different properties. Neither is a calibration measure, a deployment threshold, or a guarantee of utility after model selection.

3.2. Empirical pairwise-conflict bound

Let K criteria score all M models on image x. For model pair $p = (a, b)$, criterion k gives preference $y_{xpk} \in \{-1, 0, +1\}$. After excluding ties, let n^+_{xp} and n^-_{xp} count preferences in each direction. Any single ordering must choose one direction for p and thus disagree with at least $\min(n^+_{xp}, n^-_{xp})$ declared preferences. We aggregate the unavoidable minority count across image–pair instances:

$$B = \frac{\sum_{x,p} \min(n^+_{xp}, n^-_{xp})}{\sum_{x,p} (n^+_{xp} + n^-_{xp})}. \tag{2}$$

Criterion ties are defined by exact numerical equality; the implementation applies no tolerance. The pairwise sum is a lower bound and need not be attained by a transitive total ordering. Its denominator is the number of non-tied criterion votes, not the number of images or deployment decisions. The bound concerns complete strict model rankings. A scalar score that ties models therefore requires a deterministic tie break; unresolved scalar ties are not treated as a separate outcome. This is a preference-weighted empirical lower bound for the finite criterion set, images, and model panel. It is not the fraction of images with differing rankings, a fitted-score error, or a theorem about unobserved criteria or models. The primary estimate excludes ties; sensitivity analyses count ties as agreement or split them between directions. Confidence intervals resample images while retaining each image's complete model group, preserving within-image dependence among model-pair comparisons.

3.3. Frozen endpoint-specific scores

We enumerate single components and convex pairs on a fixed weight grid. The development selection uses COCO and VOC and requires cross-dataset consensus; the selected formulas are then frozen:

$$h_{\text{policy}} = q_{\text{CVaR}}, \qquad h_{\text{oper}} = .1q_{\text{CVaR}} + .9q_{\text{comp}}, \tag{3}$$

$$h_{\text{severe}} = .9q_{\text{CVaR}} + .1q_{\text{fp}}, \qquad h_{\text{fp}} = q_{\text{fp}}, \tag{4}$$

$$h_{\text{mixed}} = .2q_{\text{comp}} + .8q_{\text{fp}}. \tag{5}$$

For endpoint e, routing selects h_e after e has been declared. The baseline reuses h_{mixed} for all endpoints. This estimates the effect of assigning these frozen scores to these declared endpoints; it does not establish that the formulas outperform every individual metric.

Within-panel comparisons use paired image-cluster bootstrap resampling of the same image groups. Rank-endpoint draws resample jointly defined image-level paired differ-

Table 1. Evaluation panels. Records count complete model–image evaluations. COCO and VOC are used to select score formulas; ADE20K and Cityscapes are transfer evaluations.

Panel	Images	Records	Model families
COCO	4,031	24,186	FCN, DeepLab, LR-ASPP
VOC	1,449	8,694	FCN, DeepLab, LR-ASPP
ADE-S	2,000	12,000	SegFormer B0–B5
ADE-X	2,000	12,000	CNN, Swin, BEiT, M2F
City	500	3,000	SegFormer B0–B5
Total	–	59,880	24 checkpoint configurations

ences; AUROC differences use an image-clustered influence-function bootstrap. One-sided empirical p-values are $(1 + \#\{\Delta^* \leq 0\})/(B + 1)$ from 5,000 draws. The 25 panel-by-endpoint tests (five panels by five endpoints) are Holm-adjusted together at $\alpha = 0.05$, including the five mixed-failure identity comparisons; the 20 non-identity contrasts are the comparisons summarized as possible improvements. Resulting intervals are conditional on the six observed checkpoints and fixed endpoint definitions; they do not quantify variation over future datasets or architectures. Any aggregate across panels is descriptive of this evidence set, because panels are not independent draws from a defined population and ADE-S and ADE-X share images.

4. Experimental Protocol

4.1. Five locked panels

COCO val2017 [13] is evaluated through the 20 PASCAL foreground categories; only its 4,031 images with eligible components enter complete groups. All 1,449 VOC validation images [9], 2,000 ADE20K validation images [23], and 500 Cityscapes validation images [7] are used. The ADE-X panel contains two ConvNeXt/UPerNet CNNs and four transformer or mask-classification models. Model inference is serial, checkpoint revisions are pinned, and every downstream audit reads hash-locked per-image records.

Class severities and component eligibility are frozen per dataset before its evaluation. The Cityscapes policy assigns severity 5 to person, rider, motorcycle, and bicycle; severity 4 to car, truck, bus, and train. Components smaller than 16 pixels are excluded. These choices are declared experimental utilities and are not presented as validated social or physical harm weights.

4.2. Endpoints and comparisons

Policy and operational endpoints use mean per-image six-model Spearman agreement. Severe-miss, false-positive, and mixed-failure endpoints use AUROC with score loss $1 - h_e$. The scalar-conflict set contains policy utility, operational utility, false-positive quality, foreground IoU, Boundary IoU, the severity-weighted geometric aggregate, and CVaR quality. We compare the mixed-failure score with the strongest single component on each panel. Held-out results do not change the formulas in Eq. 5. The principal comparisons have deliberately different estimands. The conflict bound describes disagreement among criterion preferences for model pairs; AUROC measures discrimination between labeled image groups; and Spearman correlation measures within-image model-order agreement. We do not combine them into an omnibus score. The mixed-failure AUROC comparison uses the fixed high-false- positive threshold of 0.25. The strongest single component is selected within each panel for this descriptive comparator; it is not an independent external baseline. Selecting the maximum single-component AUROC on the same panel makes that comparator optimistic and the reported mixed-minus-best-single gap conservative with respect to this component set.

4.3. Uncertainty and robustness

Primary within-panel intervals use 5,000 paired image-cluster bootstrap resamples. Endpoint-routing comparisons are corrected jointly across five panels and five endpoints with Holm's procedure. Equal-panel syntheses use 20,000 panel bootstrap resamples and leave-one-panel-out means. Ranking stability uses 5,000 image bootstraps and exhaustively evaluates every three-, four-, and five-model subset. The post-confirmation failure-definition audit evaluates seven false-positive thresholds from 0.10 to 0.50 using 3,000 clustered resamples per panel–threshold cell. This threshold audit and the Cityscapes class-presence analysis are explicitly exploratory.

For endpoint-routing uncertainty, each bootstrap draw resamples images as clusters and carries all six model records for a sampled image together. The within-panel contrast is computed from paired routed and mixed scores in each draw. Holm correction controls familywise error over 25 panel-by-endpoint tests; the identity comparison for the mixed-failure endpoint is an identity, not evidence of improvement. Equal-panel means and panel ranges summarize only these five existing panels. We do not interpret a bootstrap over five panels as population-level inference.

The development chronology is material. Score formulas are selected on COCO/VOC; ADE-S evaluates dataset transfer, ADE-X evaluates transfer across architectures, and Cityscapes evaluates a structured driving domain. Conflict, mechanism, routing-synthesis, and class-presence analyses were designed after the initial confirmations. We report them as post-confirmation analyses and do not use them to retune the selected formulas.

The two ADE20K panels are not independent dataset replications: they use the same validation images but different checkpoint families. Inference about architecture transfer is consequently conditional on this fixed image

Table 2. Empirical conflict bound B and mixed-score AUROC difference relative to the strongest single metric at the fixed 0.25 false-positive threshold.

Panel	B [95% CI]	Mixed − best single
COCO	.160 [.157,.162]	+.068
VOC	.142 [.139,.146]	+.067
ADE-S	.163 [.159,.166]	+.026
ADE-X	.186 [.183,.189]	+.028
City	.106 [.102,.111]	−.124

Table 3. Criterion-set sensitivity of the empirical conflict bound on the four panels with a common subset audit. The first score column uses policy utility, operational utility, and false-positive quality. The middle column adds foreground IoU and Boundary IoU. The final column further adds the severity-weighted component aggregate and CVaR quality. Values are point estimates; primary uncertainty intervals are in Table 2.

Panel	Policy utility, operational utility, false-positive quality	Plus foreground and Boundary IoU	All criteria in study
COCO	.133	.163	.160
VOC	.120	.145	.142
ADE-S	.060	.146	.163
ADE-X	.061	.175	.186

set. Cityscapes comprises 500 validation images, and class-presence strata can be sparse; all class-specific estimates are exploratory. We report image counts, not only model–image record counts, because records from different models share image clusters.

5. Results

5.1. Criterion preferences conflict across panels

The bound is positive in all five panels (Table 2, Fig. 1) and is largest in the architecture-diverse ADE-X panel. On the four panels other than Cityscapes, leave-one-criterion-out, model-subset, image-subsample, and split-half analyses preserve this result. Across all three-model subsets in ADE-X, the bound ranges from 0.169 to 0.202; the median split-half criterion-pair conflict correlation is 0.986. Counting ties as agreement lowers the estimate, whereas splitting ties raises it; the bound remains positive under both alternatives.

The numerical range should be read against the definition in Eq. 2: it is a lower bound on disagreement over criterion votes, not a claim that a fraction of deployment decisions will be wrong. In the four-panel sensitivity audit, the bound remains positive for deployment-only and conventional-plus-deployment subsets (Table 3). It is smaller for deployment-only criteria on ADE20K (0.060–0.061) than for the five-criterion subset (0.146–0.175). The effect is therefore not restricted to including only tail-oriented quality measures, but its magnitude depends materially on which criteria are declared.

Table 4. Endpoint-specific score minus reuse of the mixed-failure score. Mean is the unweighted mean of the five panel estimates; range is the observed minimum–maximum panel estimate, not a confidence interval.

Endpoint	Mean [range]	>0/5	LOO min.
Policy rank	.318 [.067,.589]	5	.251
Operational rank	.138 [.071,.175]	5	.129
Severe-miss AUROC	.237 [.138,.365]	5	.205
False-positive AUROC	.0065 [0,.014]	4	.0046

At the fixed false-positive threshold, the mixed-failure score improves AUROC over the strongest single component by 0.026–0.068 on four panels. On Cityscapes, it is lower than severity-weighted arithmetic by 0.124 (95% clustered CI for the mixed-score difference: $[-.194, -.061]$). The direction and size of the observed difference therefore vary across panels.

5.2. Sensitivity to the failure definition

Across seven false-positive thresholds, the mixed score exceeds the strongest single metric in 7/7 VOC cells and is lower in 7/7 Cityscapes cells. The difference is positive in 5/7 COCO cells and 4/7 cells on each ADE panel (Fig. 2). The measured advantage thus depends on the dataset and the operational definition of failure. These threshold-sweep results were obtained after the initial confirmations and are treated as sensitivity analysis.

5.3. Endpoint-specific scores

Endpoint-specific scoring increases policy-rank agreement, operational-rank agreement, and severe-miss AUROC on each panel; the corresponding leave-one-panel-out means are also positive (Table 4). Nineteen of the 20 non-identity panel–endpoint contrasts are significant after the 25-test Holm correction; the five mixed-failure identity comparisons remain in the correction family but are not interpreted as improvements. The remaining comparison, false-positive AUROC on Cityscapes, is an exact tie at AUROC 1.0. On Cityscapes, endpoint-specific scoring changes policy agreement by 0.067, operational agreement by 0.071, and severe-miss AUROC by 0.138. Because the mixed-failure endpoint uses the mixed-failure score by definition, that identity comparison is omitted.

Figure 3 exposes the paired image-level uncertainty and heterogeneity hidden by equal-panel means. Effects are positive for policy, operational, and severe-miss endpoints in each panel; false-positive scoring is an exact tie on Cityscapes, where AUROC is already 1.0. The panel ranges are descriptive, not predictive intervals for new domains.

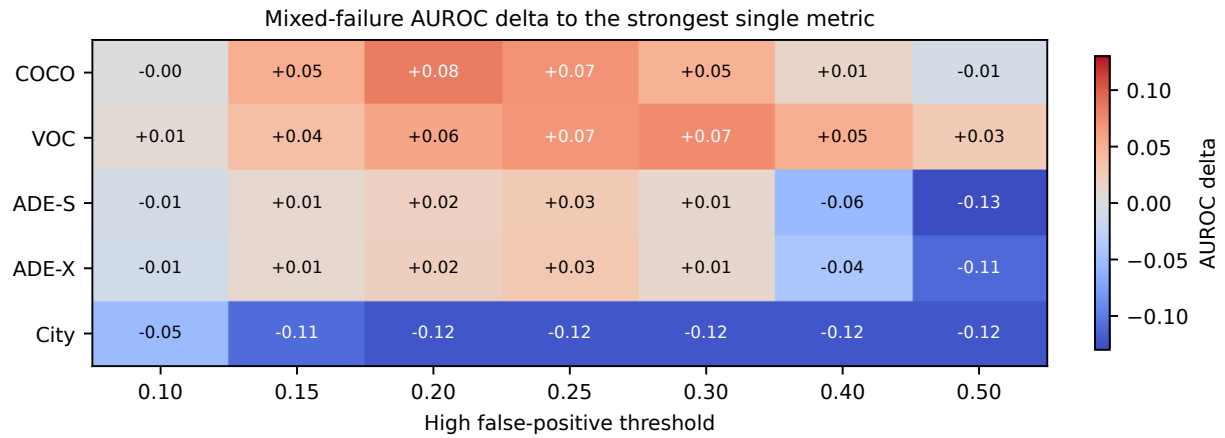


Figure 2. Post-confirmation sensitivity of mixed-failure AUROC. Each cell gives the frozen mixed score minus the strongest single metric at the indicated false-positive threshold. The difference changes sign on COCO and both ADE panels, remains positive on VOC, and remains negative on Cityscapes.

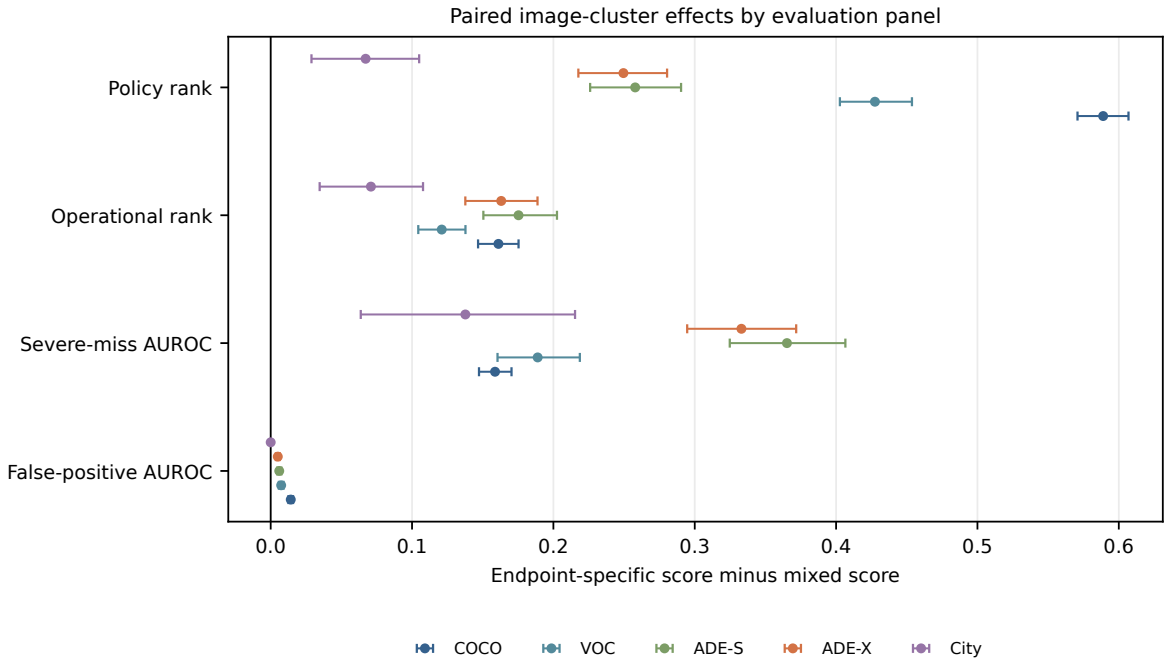


Figure 3. Paired endpoint-routing effect by panel. Dots are the within-panel routed score minus the mixed score; horizontal intervals are 95% paired image-cluster bootstrap intervals. Each interval is conditional on the six evaluated checkpoints, image set, and endpoint labels for that panel. The five panel effects are shown separately; they are not independent draws from a population of datasets.

Within-panel intervals condition on the checkpoint panel and specified target labels.

These results compare endpoint-specific scoring with reuse of one mixed-failure score. They do not show that every selected formula exceeds every individual metric: the policy score is CVaR, the false-positive score is the direct false-positive measure, and severity-weighted arithmetic slightly exceeds the selected severe-miss score on Cityscapes.

5.4. Cityscapes failure composition

Among the 3,000 Cityscapes model–image records, 2,763 are severe-only failures, six contain both failure types, one is high-false-positive only, and 230 contain neither. The mixed score assigns weight 0.8 to the false-positive com-

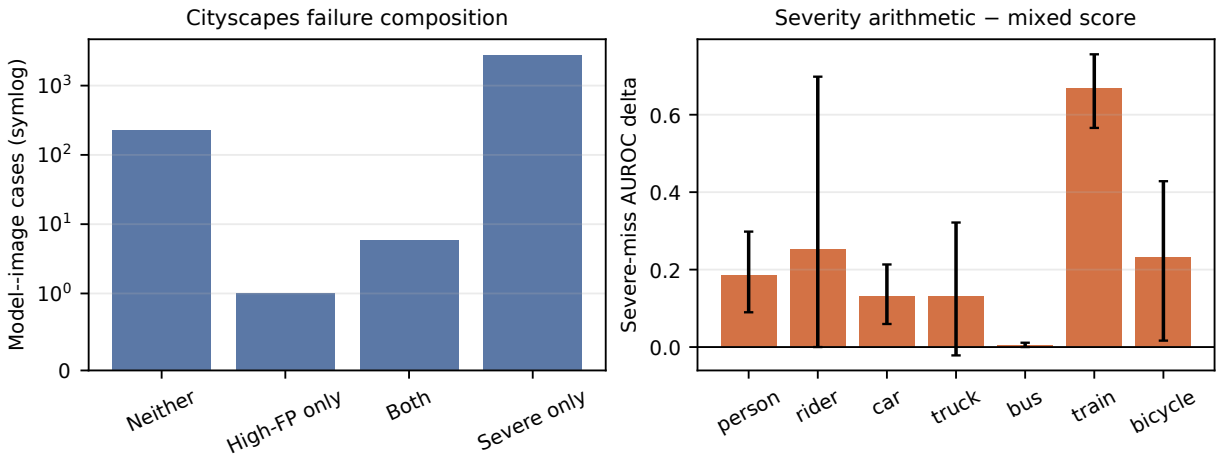


Figure 4. Post-confirmation Cityscapes analysis. Left: severe-only failures dominate the fixed mixed endpoint. Right: within scenes containing each safety-critical class, severity arithmetic minus mixed-score severe-miss AUROC (95% image-cluster bootstrap confidence intervals). Motorcycle is omitted because its stratum contains 93 images (558 model–image records), all positive for severe miss, leaving no negative records and making AUROC undefined. Class presence defines a stratum and does not identify which class caused an image-level miss.

ponent, a failure mode that is rare in this panel. In class-presence strata, severity arithmetic exceeds the mixed score in severe-miss AUROC by 0.185 for person scenes (95% CI [.090,.298]), 0.231 for bicycle scenes ([.017,.428]), and 0.131 for car scenes ([.060,.213]). Intervals for rider, truck, and bus touch or cross zero; the small train subset does not support a broader conclusion.

This breakdown is consistent with a possible mechanism for the negative aggregate comparison: the mixed score assigns 0.8 weight to false-positive quality, while false-positive-only failures are rare under the panel's fixed label definition. It does not establish that the weighting caused the regression, nor that severity arithmetic is preferable under other class costs. Class presence is not per-class causal attribution, and the stratified comparisons are exploratory rather than multiplicity-adjusted confirmatory tests.

5.5. Aggregate ranking stability

Across panels, aggregate mixed-score rankings have Spearman correlations of 0.943–1.0 with aggregate policy utility; all lower confidence limits are at least 0.829. Top-model selection is less stable: on the two ADE panels, bootstrap top-model agreement is 0.614–0.680. Aggregate ranking agreement therefore coexists with uncertainty about which model ranks first and with the per-image criterion conflicts reported above.

5.6. Reproducibility

The prepared artifact package includes a figure generator that reads aggregate JSON metrics for E021, E023, E024, E025, E027, and E028; configuration snapshots and tabular aggregates support inspection of these analyses. The run artifacts record data/profile hashes, model revisions, seeds, software environment, and execution status where applicable. The package excludes licensed source data, checkpoints, and per-image Cityscapes records; it therefore supports figure regeneration and aggregate-result inspection, but not an independent rerun of every image-level analysis. E021, E022, and E024–E028 are post-confirmation analyses without new model inference; E023 is the Cityscapes confirmation evaluation. These statements describe supplied run artifacts, not a claim that every upstream input is redistributed.

6. Limitations

Scope of the conflict bound. Equation 2 applies to the seven declared criteria, evaluated images, and model panels. It is an empirical bound, not a general impossibility result. Alternative criteria or task-specific utilities may change its value. The primary analysis excludes ties; two alternative tie conventions are also evaluated.

Endpoint construction. Policy utility, severe-miss labels, CVaR, and the component scores share component-coverage measurements. They are therefore not independent external safety outcomes. Endpoint-specific scoring measures agreement with these declared quantities; it does not estimate human welfare or causal risk. The severity maps are experimental assumptions and have not been validated as social or physical harm weights. The conflict bound is likewise criterion-dependent and preference-weighted over non-tied votes; it is not an estimate of human disagreement,

deployment loss, or error from a learned evaluator. We lack independent expert labels or prospective decisions to validate these utilities.

Post-confirmation analyses. COCO and VOC are used to select score formulas, and ADE20K and Cityscapes are held-out transfer evaluations. The conflict sensitivity analysis, routing synthesis, threshold sweep, and class-presence analysis were designed after the initial confirmations. They are identified as post-confirmation analyses and were not used to retune the formulas.

Panel coverage. The study covers four datasets and multiple architecture families, but each panel contains six models. Image bootstrap intervals quantify uncertainty conditional on those models and do not establish behavior on unseen architectures. ADE-S and ADE-X use the same images, so the five panels are not five independent datasets. The panels are convenience evaluations assembled from the available locked artifacts. Only ADE20K and Cityscapes are transfer datasets beyond COCO/VOC, and the ADE panels share images. Confidence intervals quantify image resampling conditional on fixed checkpoints; they do not establish population-level generalization. Positive panel-wise results apply to these specific settings, not to an inferred population of datasets or architectures.

Selection and multiplicity. Formula selection uses two development datasets and then freezes the selected scores, which reduces but does not eliminate selection optimism. The strongest-single comparator is selected separately within each evaluation panel and is consequently descriptive, not an independent external baseline. Mechanism and sensitivity analyses designed after initial confirmations are labeled exploratory and should motivate, not replace, a preregistered independent replication.

Cityscapes class analysis. Severe misses are prevalent under the selected component threshold. As a result, motorcycle-stratum AUROC is undefined and several class-specific intervals are wide. The analysis stratifies by scene-level class presence; it cannot identify which class caused an image-level miss.

Use in model selection. Endpoint routing requires a user to specify the decision objective. If that objective is misspecified, the selected score will faithfully evaluate the wrong target. Segmentation quality, boundary quality, calibration, resource cost, and domain-specific safety constraints should therefore be reported separately when relevant. Reliance on any single score can also encourage optimization to that score rather than to the intended application.

7. Conclusion

Across five fixed evaluation panels, the declared seven criteria induce pairwise preference conflict: the empirical lower bound ranges from 10.6% to 18.6%. This bound is conditional on the evaluated criteria, model groups, images, and tie rule. It is not evidence that a corresponding fraction of real-world decisions will be wrong. The mixed-failure score improves failure-detection AUROC on four panels but is lower than severity-weighted arithmetic on Cityscapes and does not preserve policy or operational rankings. The results reject a universal-win interpretation of the mixed score.

The practical implication is procedural rather than a new default metric. Evaluation should name the decision endpoint, specify its utility or failure definition before examining results, report conventional segmentation metrics alongside endpoint-specific measures, and validate any selected score on independent data with uncertainty intervals matched to the sampling unit. When endpoints conflict, that conflict should be reported rather than hidden inside a single aggregate. Our frozen routing comparison is evidence for this reporting discipline within the studied panels; it is not evidence that the particular score formulas are optimal for deployment.

An independent, preregistered study with externally elicited utilities, additional datasets and architectures, and prospective decision outcomes is needed to determine whether these empirical trade-offs generalize. Until then, the defensible conclusion is narrow: failure-detection discrimination, model-ranking agreement, and deployment utility are distinct evaluation targets and should not be treated as interchangeable without validation.

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